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SUBJECT CODE NO: - RNP-8059
FACULTY OF SCIENCE AND TECHNOLOGY
F.E (Group A) (NEP) Sem - II
Examination May/June-2026
Basics and Mechanical

[Time: 3:00 Hours]

[Max. Marks: 60]

Please check whether you have got the right question paper.

N. B

- 1) Use of non-programmable calculator is allowed
- 2) Q.no.1 is compulsory
- 3) Solve any four questions from Q. nos 2, 3, 4, 5, 6 and 7

Q.1 Attempt any 10 questions from the following.

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1. Which of the following is NOT a mode of heat transfer?
 - a) Conduction
 - b) Convection
 - c) Radiation
 - d) Reflection
2. In which process does no heat enter or leave the system?
 - a) Adiabatic Process
 - b) Isothermal Process
 - c) Isochoric Process
 - d) Isobaric Process
3. In a steady-state process, the thermodynamic properties of the system:
 - a) Change over time
 - b) Remain constant
 - c) Depend on external factors
 - d) Can only increase
4. Which of the following instruments is used to measure temperature?
 - a) Manometer
 - b) Thermometer
 - c) Calorimeter
 - d) Pyrometer
5. Which of the following is a major function of a compressor in a refrigeration system?
 - a) To expand the refrigerant
 - b) To cool the refrigerant
 - c) To compress the refrigerant
 - d) To increase the volume of the refrigerant
6. Which of the following materials is commonly used in the construction of machine tools?
 - a) Aluminum
 - b) Steel
 - c) Plastic
 - d) Glass
7. Which of the following is the function of a key in power transmission?
 - a) To connect rotating parts
 - b) To lock the motor
 - c) To change the gear ratio
 - d) To store energy
8. In a refrigeration system, the expansion valve is used to:
 - a) Increase pressure
 - b) Increase temperature
 - c) Reduce pressure
 - d) Increase work output

9. Which of the following metal joining processes involves melting the base material and adding a filler material to form a strong bond?
 - a) Soldering
 - b) Welding
 - c) Riveting
 - d) Brazing
10. In which of the following processes is a material deformed to achieve a desired shape using compressive forces?
 - a) Forging
 - b) Casting
 - c) Electroplating
 - d) Grinding
11. Which of the following statements about CNC (Computer Numerical Control) machines is true?
 - a) CNC machines require manual control for all operations
 - b) CNC machines are used to produce simple designs with limited precision
 - c) CNC machines use computer programs to control the movement of the tool
 - d) CNC machines are more cost-effective than conventional machines for all operations
12. Which part of the lathe machine is responsible for holding the work piece during operation?
 - a) Tailstock
 - b) Chuck
 - c) Tool Post
 - d) Cross-slide

- Q.2** a) State the various modes of heat transfer and explain them briefly. **5**
 b) Write short note on forms of energy. **5**
- Q.3** a) Explain thermodynamics equilibrium in details. **5**
 b) What is pressure? Explain its anyone type of pressure measuring device. **5**
- Q.4** a) What is boiler? Explain in detail any one. **5**
 b) Explain 4 stroke SI engine. **5**
- Q.5** a) State and explain the selection criteria for Engineering Materials. **5**
 b) What is heat treatment process? Explain in detail annealing heat treatment process. **5**
- Q.6** a) Explain the surface grinding machine with neat sketch. **5**
 b) Explain CNC in Details. **5**
- Q.7** a) Differentiate between flat and v belt drive. **5**
 b) Explain the construction and working of single plate clutch with neat diagram. **5**